

Unit-5 One Shot + 3 PYQs



Question 1 -

Unit - 5

Q.1 Explain Cycle stealing?

Ans Cycle stealing allows DMA controllers to transfer one data word at a time after which it must return control of the buses to the CPU.

→ The CPU usually delays its operation for one memory cycle to allow the direct memory I/O transfer to "steal" one memory cycle.

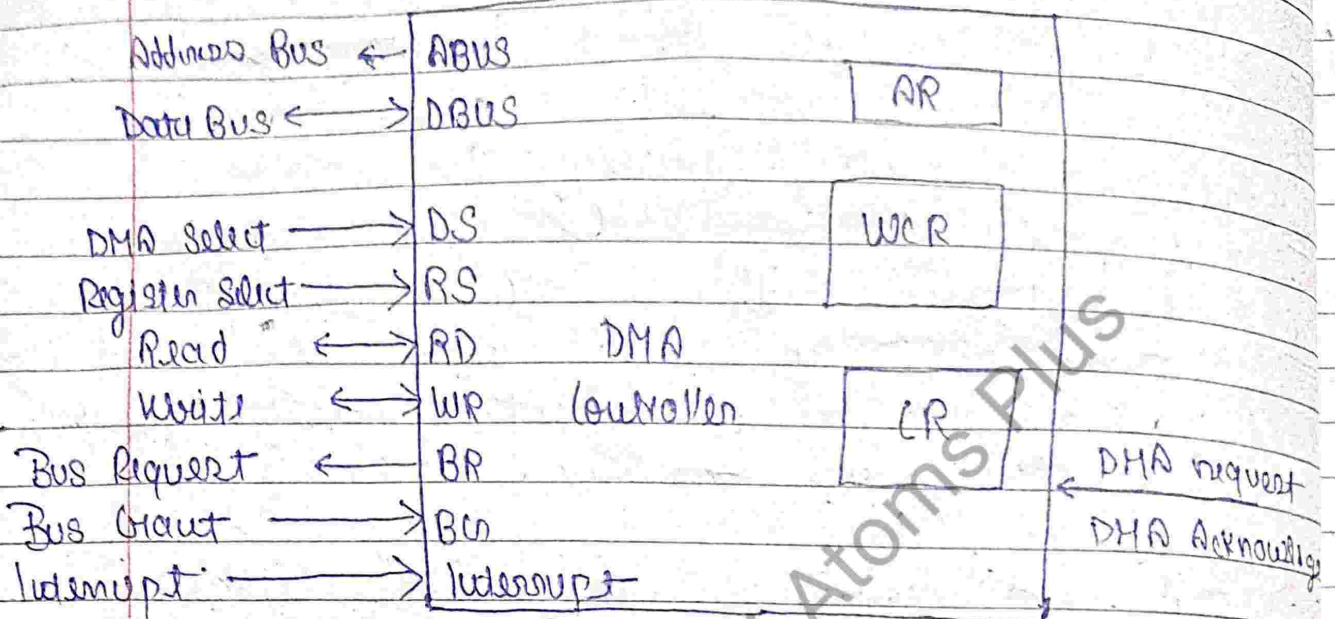
⇒ This mode is used in slow devices such as keyboard.

Q.2 Block diagram of DMA controller ^{7 marks}

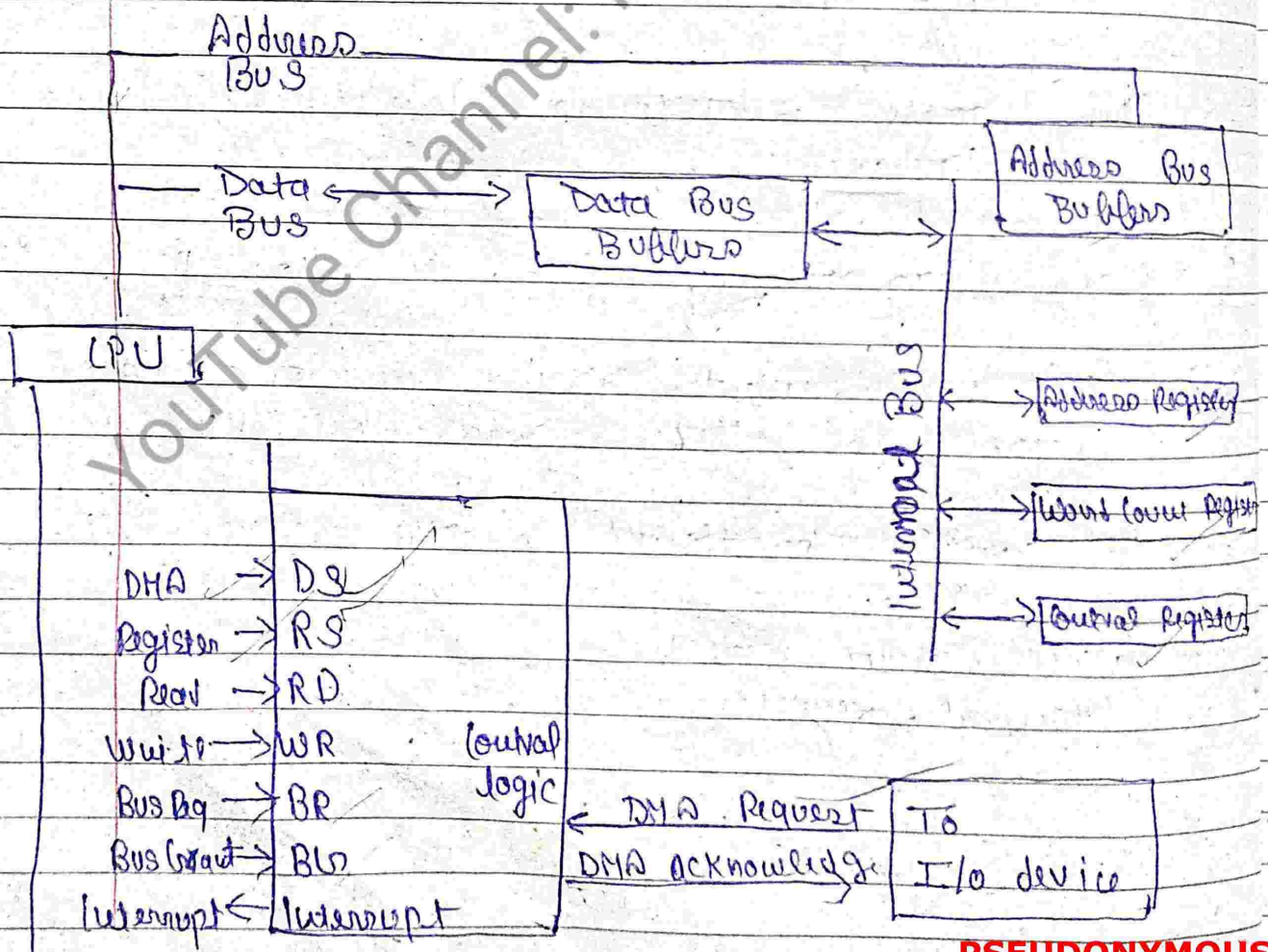
Ans A device which allows direct memory access is called DMA controller.

⇒ The registers mentioned in DMA controller are

- (i) Address register
- (ii) Word count register
- (iii) Control register



Block diagram of DMA controller:-



① The DMA controller holds the usual circuits of an interface to communicate with the CPU & I/O devices.

② The DMA controller has three registers.

- 1] Address Register
- 2] Word Count Register
- 3] Control Register

(i) Address Register:- It contains an address to specify the desired location in memory. It is incremented by 1 after each word is transferred to memory.

(ii) Word Count Register:- It holds the number of words to be transferred. It is decremented by one after each transfer & internally tested for zero.

(iii) Control Register:- It specifies the word of transfer i.e. Read/Write the memory.

③ The DMA controller communicates with the CPU via data bus & ^{address} bus (lines).

④ The registers in DMA are selected by the CPU. Through the register in DMA are selected by the CPU through the address bus by enabling DS (DMA select) & RS (Register Select) inputs.

- The RD and WR (read & write) are bidirectional for the CPU.
- When Bh (Bus master) I/P is 0, the CPU can communicate with the DMA registers through the data bus to read from one write in the ~~DMA~~ registers peripheral device.
- when $Bh = 1$, the CPU has relinquished the bus & the DMA can communicate directly with memory by specifying an address in the address bus & activating RD or WR control.
- DMA communicate with the external I/O devices through the request & acknowledge line by using handshaking procedure.

Q.IMP What do you mean by asynchronous and data transfer? Explain strobe control and hand shaking mechanism.

Ans It is used when speed of I/O devices do not match with processor & Timing characteristics of I/O devices is not predictable.

Def! — Asynchronous data transfer between two independent units requires control signals to be transmitted between communicating units to indicate the time at which data is being transmitted.

In Asynchronous Data Transfer,

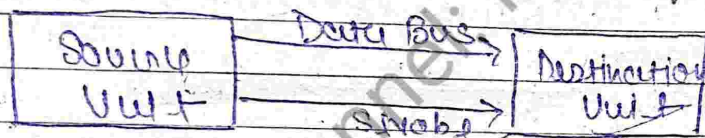
2 type of method are used :-

- 1] Strobe Control.
- 2] Handshaking Method.

⇒ Strobe Control :-

- It employs a single control line to time each transfer.
- The strobe may be activated by either source or destination unit.

Source - initiated strobe for data transfer :-



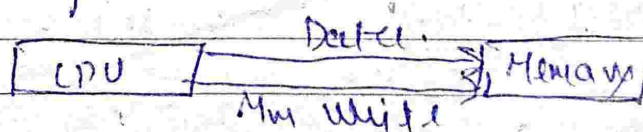
(a) Block Diagram



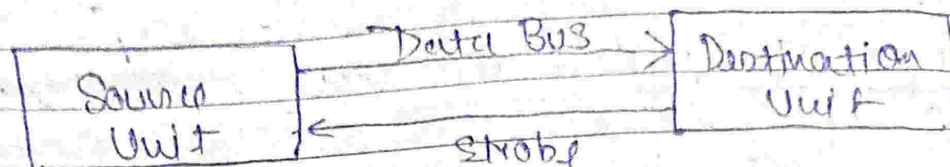
Strobe

(*) Timing Diagram

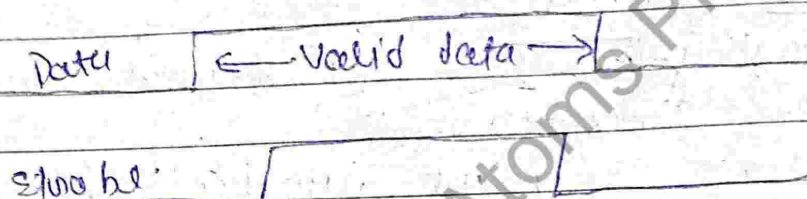
eg Memory write control signal from CPU to memory unit :-



Destination - Initiated Strobe for data transfer

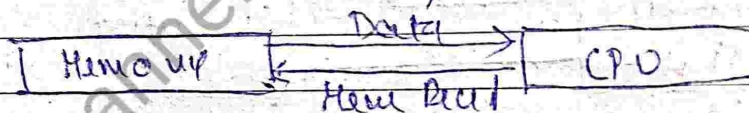


(a) Block diagram



(b) Timing diagram

Eg Memory read control signal from CPU to memory unit



=> Disadvantage

-> Source unit that initiates the transfer has no way of knowing whether the destination unit has actually received the data item that was placed in the bus

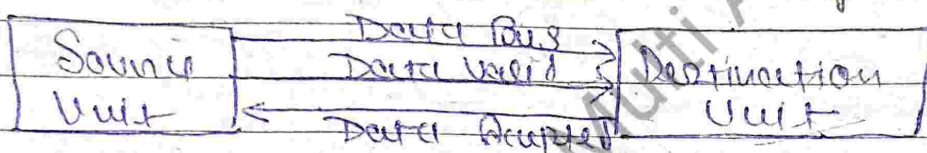
-> Similarly Destination unit that initiates the transfer ~~data~~ has no way of knowing whether the source unit has actually placed the data on the bus

② Handshaking Method :-

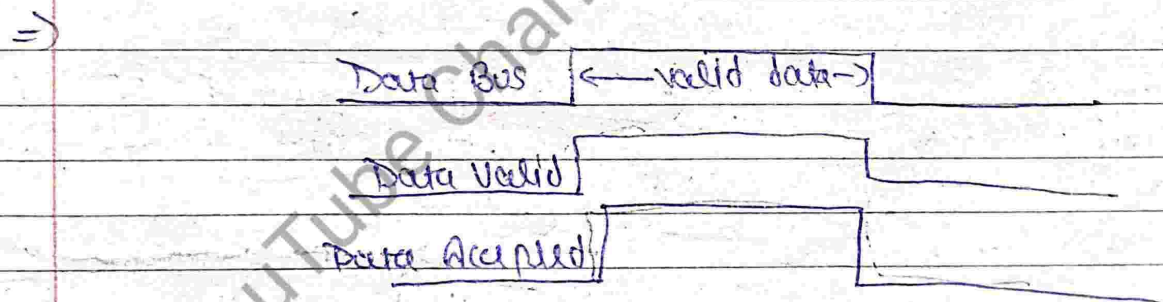
→ Solves the problem strobe method by introducing a second control signal that provides a reply to the unit that initiates the control.

→ Strobe (control Signal) + Acknowledgement Signal

⇒ Source - initiated Handshaking :-



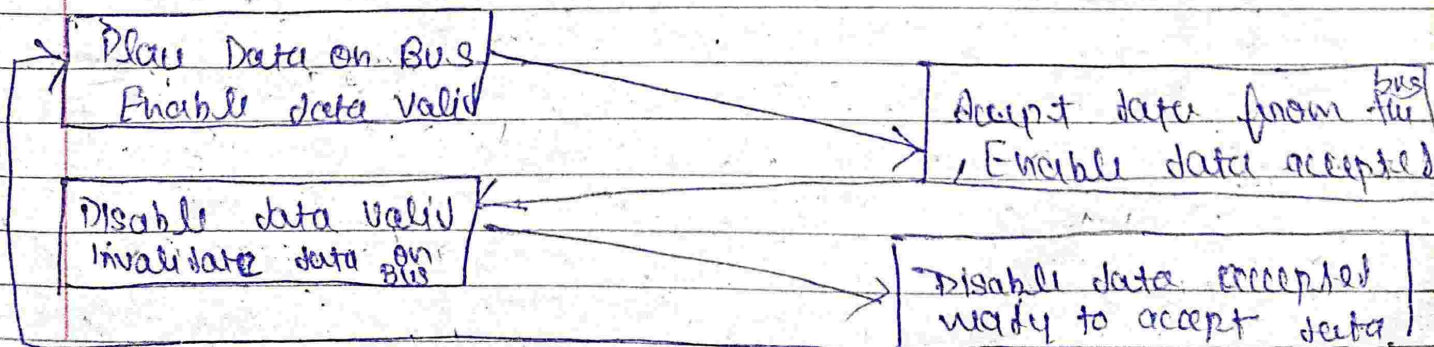
⊗ Block diagram



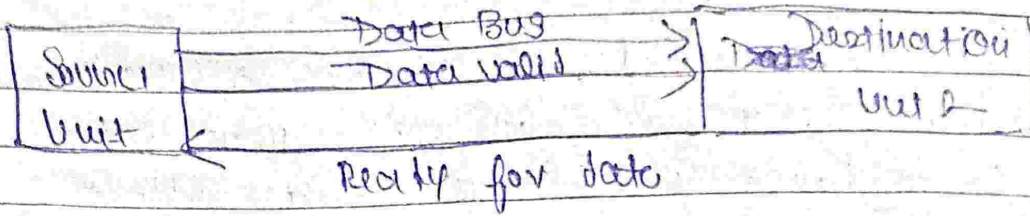
Ⓐ Timing Diagram

Source

Destination



Destination - Initiated Handshaking



(*) Block Diagram

Ready for data

Data Valid

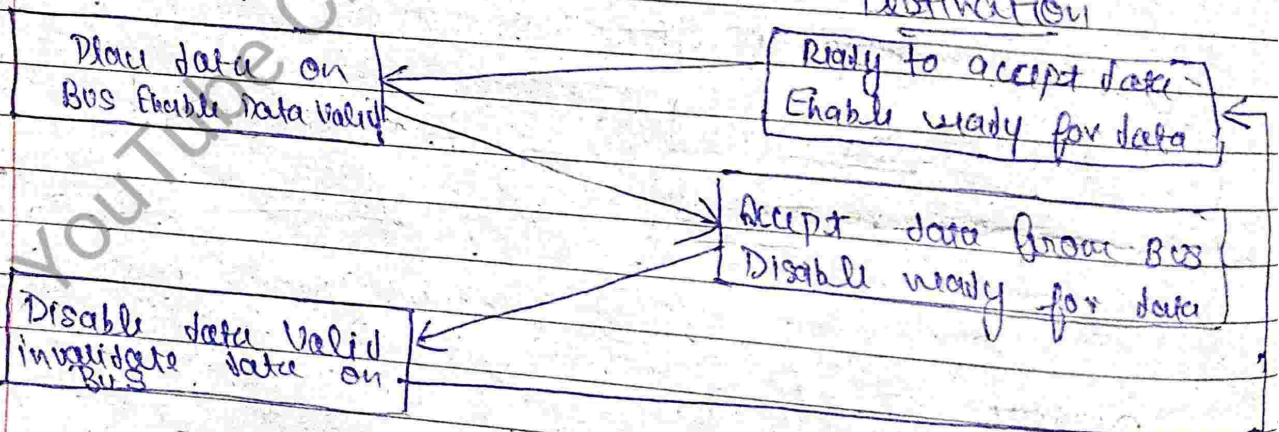
Data Bus

Valid data

(*) Timing Diagram

Source

Destination



7 marksQ4Discuss the mode of data transfer? 4 marks

Ans Data transfer between the central computer & I/O devices may be handled in a variety of modes.

- 1] Programmed I/O
- 2] Interrupt Initiated I/O
- 3] DMA

⇒ Programmed - I/O

→ Programmed I/O operations are a result of I/O instructions written in computer program.

⇒ Each data transfer is initiated by an I/O instruction in the program (to access register memory on a device).

⇒ Transferring data under program control requires constant monitoring of the I/O devices by the CPU.

⇒ CPU make a request & then CPU stays in program loop (polling) until the I/O device indicates that it is ready for data transfer.

Interrupt Initiated I/O :-

- ① In interrupt - initiated I/O instead of continuous monitoring of I/O, interface will be informed to ISSUE an interrupt request signal when data are available from the device.
- ② Meanwhile, CPU proceeds to execute another program & interface keep monitoring the device.
- ③ When device is ready for data-transfer it generates interrupted request.
- ④ Upon detecting the external interrupt signal by the CPU stops the task it is performing, process the I/O data-transfer & then resumes the original task it was performing.

DMA

Date			
Page No.			

Direct Memory Access

- To transfer large block of data of high speed between external and main memory DMA approach is used.
- ⇒ DMA allows data transfer directly between I/O device & main memory with minimal intervention of CPU.
- ⇒ DMA means CPU grants I/O interface authority to read from or write to memory without involvement.
- ⇒ DMA itself control data transfer between main memory & I/O devices.
- ⇒ CPU is involved in beginning & end of the transfer & interrupted only after entire block has been transferred.
- ⇒ CPU asks the DMA controller to transfer data between a device & main memory & then CPU proceeds to execute other tasks.
- ⇒ The DMA controller issues a requests to the I/O device, waits & manages data transfer between the device & main memory.
- ⇒ When data transfer is finished, the DMA interrupt the CPU.

Q Explain destination-initiated, transfer using Handshaking method.

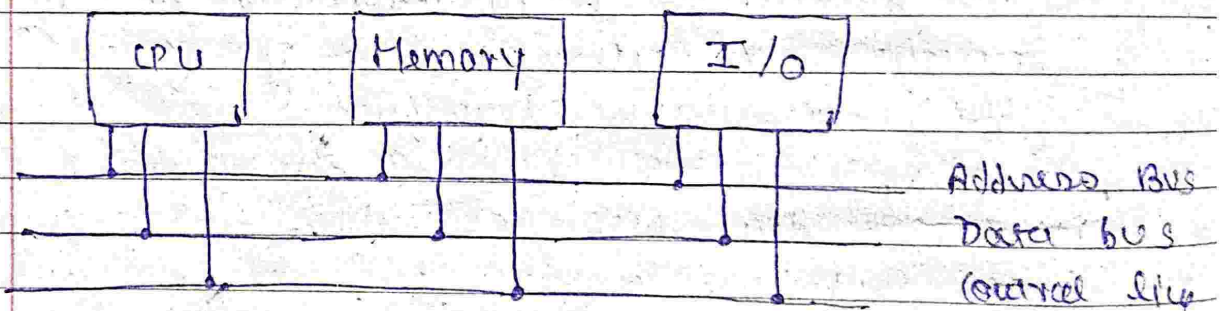
Sol Refer Q.3.

Q. 1.1.2 Different method of asynchronous data transfer.

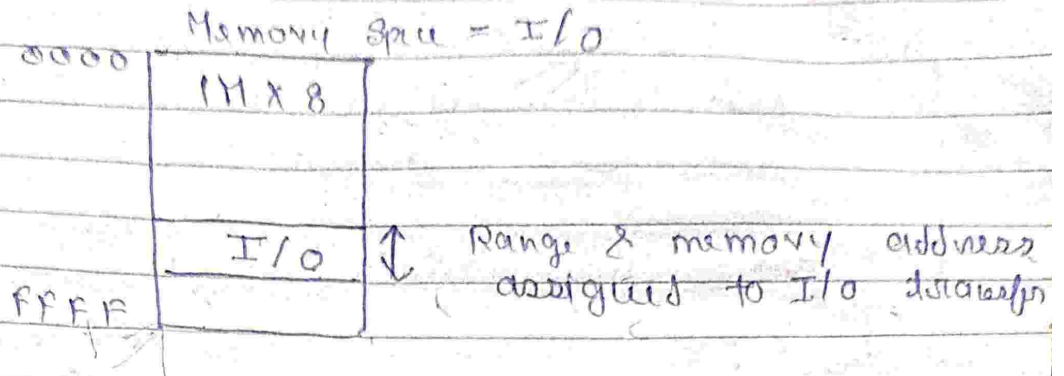
Sol Refer Q.3.

Q Explain how the computer buses can be used to communicate with memory & I/O. Also draw the block-diagram for CPU - I/O communicate.

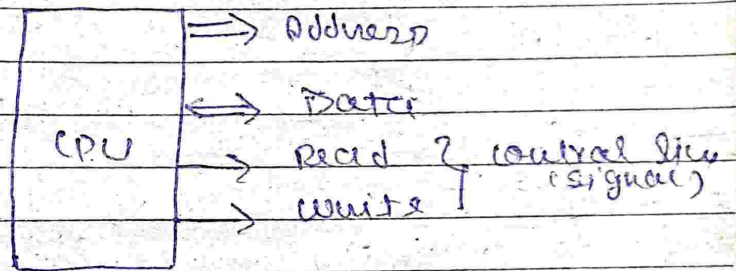
Sol (a) Memory Mapped I/O



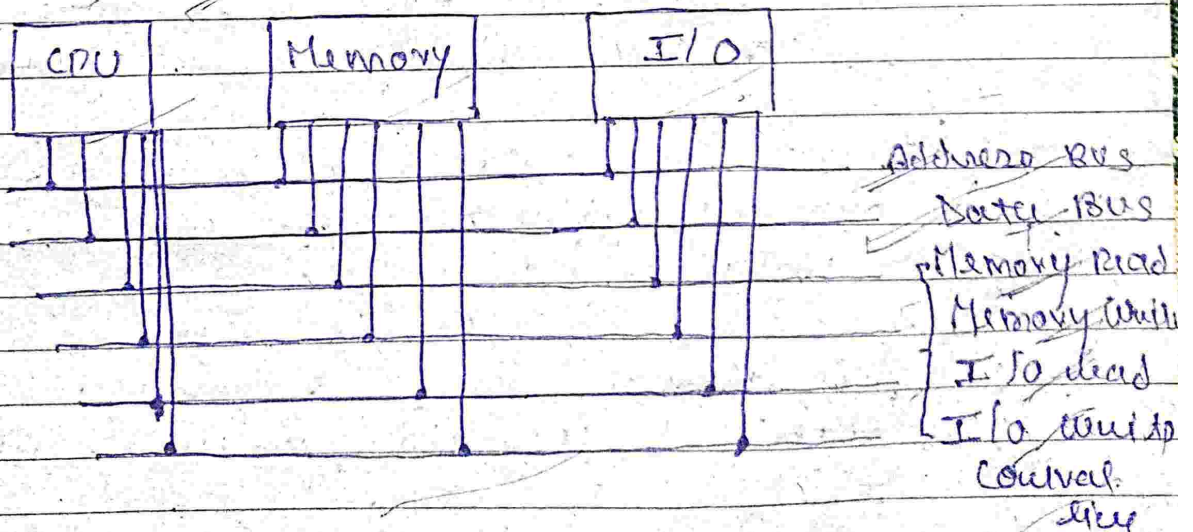
- ① Address, data & control bus are common for I/O devices & memory.
- ② I/O devices & memory share same address space due to which addressing capability of memory become less because some part is occupied by the I/O.



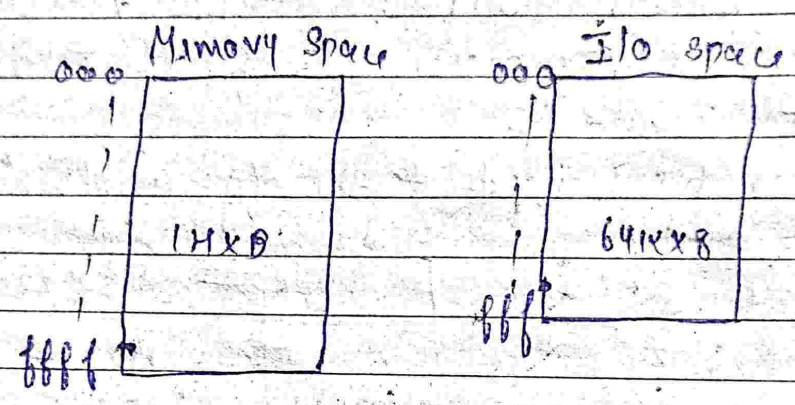
- ① Single set of read/write control lines for I/O & memory transfer.
- ② I/O transfer looks memory read/write.
- ③ No special commands for I/O transfer for (IN & out).
- ④ MOV can be used to perform both I/O memory transfer.



⇒ (b) Isolated I/O

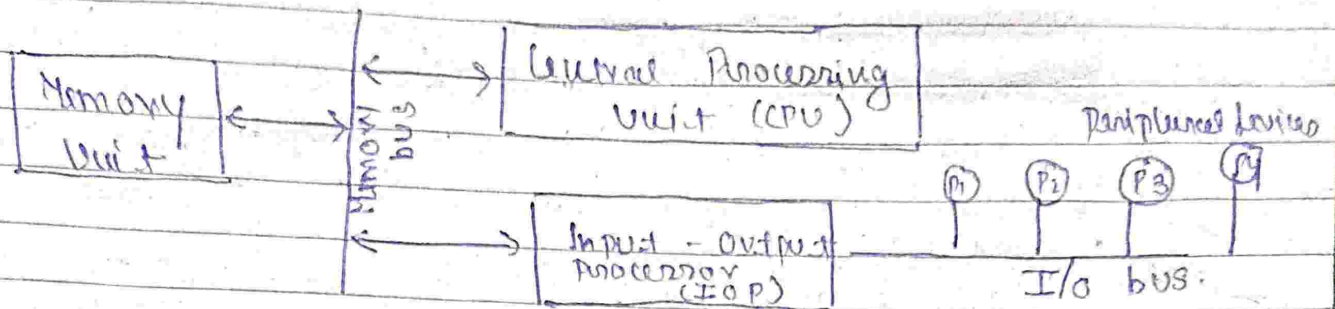


- ① The address & data bus are common but separate read & write control line for I/O and memory
- ② I/O device & memory both a separate address space.
- ③ Separate read/write control line for I/O & memory transfer.



- ④ I/O read and I/O write for I/O transfer
- ⑤ Memory read & write for memory transfer
- ⑥ Special command is used for I/O transfer
IN & OUT instructions for memory transfer
Mov instruction
- ⑦ Because it is isolated I/O address from memory address that why called isolated I/O.
- ⑧ The address for I/O here is called port, hence also called port-mapped I/O.

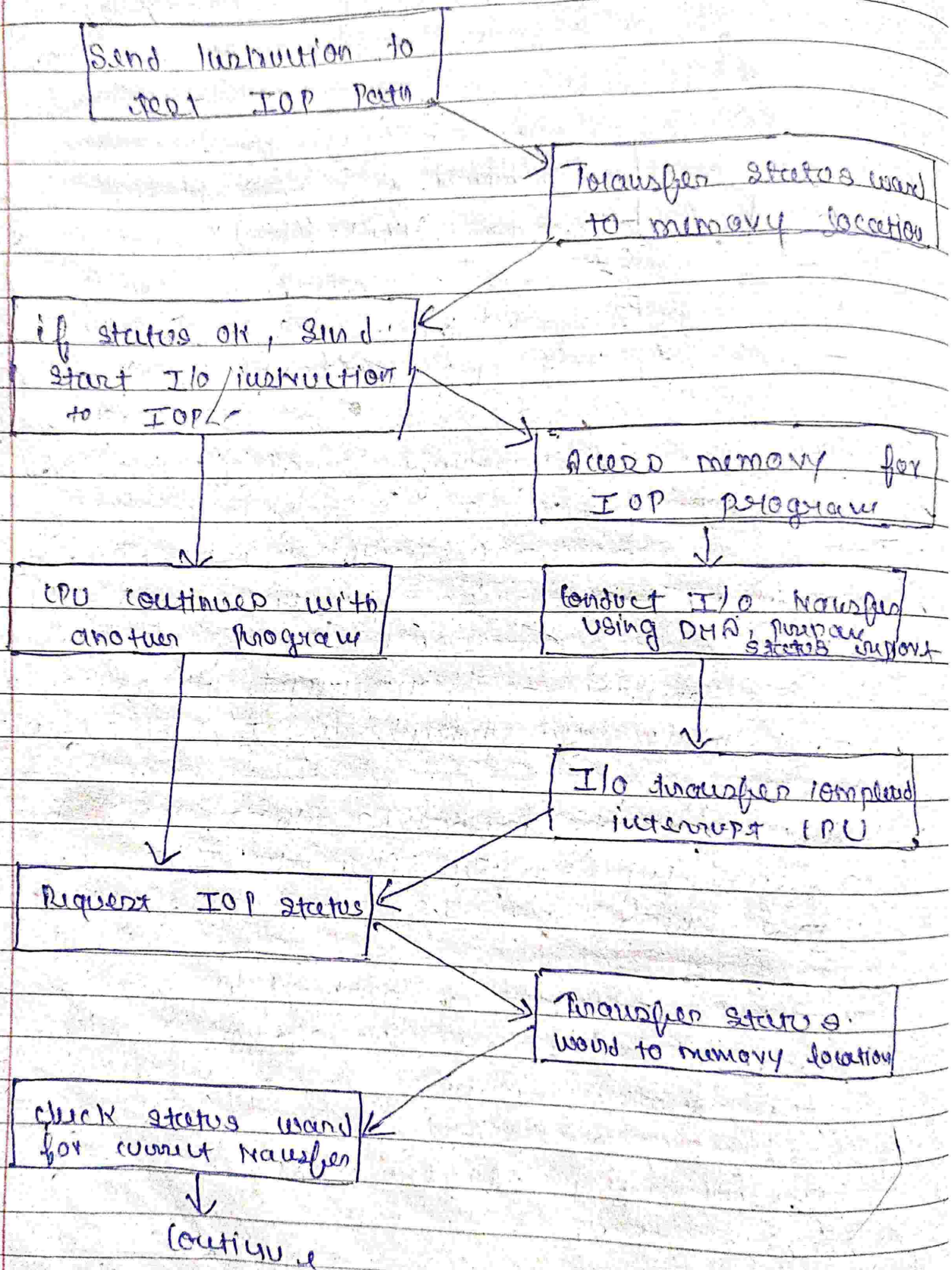
Input - Output Processor (IOP)



block diagram of a computer with IOP

IOP is a processor with direct memory access capability that communicate with I/O devices.

- It is similar to CPU except that it is designed to handle the detail of I/O processing.
 - Unlike the DMA controller that must be set up virtually by the CPU, the IOP can fetch and execute its own instructions.
 - The IOP provides a path for transfer of data between various peripheral devices and the memory unit.
- ⇒ The CPU only initiates the I/O program, after that IOP operates independently of the CPU.



Q. What is I/O control method.
 Sol. Refer to Q. 1.

Q. ^{Imp} Write down the diff. b/w isolated I/O and memory mapped I/O. also discuss advantages & disadvantages.

Sol. Refer to Q. 2.

⇒ Advantages of Isolated :-

① The devices of I/O are directed in a separate domain as compared to memory.

② A total of 1MB address space is allowed for memory applications.

③ In order to maximize the I/O operations separate instructions are always provided to perform these operation.

⇒ Disadvantages :-

1. The data transfer only occurs between the I/O port and the registers.

⇒ Advantages of memory mapped I/O :-

1. I/O intensive operation is fast.

2. The OS like library need less ROM.

⇒ Disadvantages of memory mapped I/O :-

1. Performance is reduce by the use of memory mapped I/O.

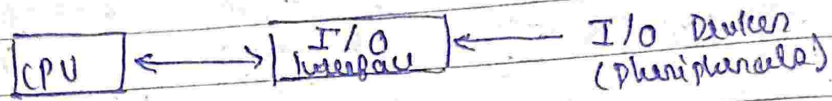
2. If an I/O error on a memory mapped file cannot be caught by the application, **PSEUDONYMOUS**

Q Imp

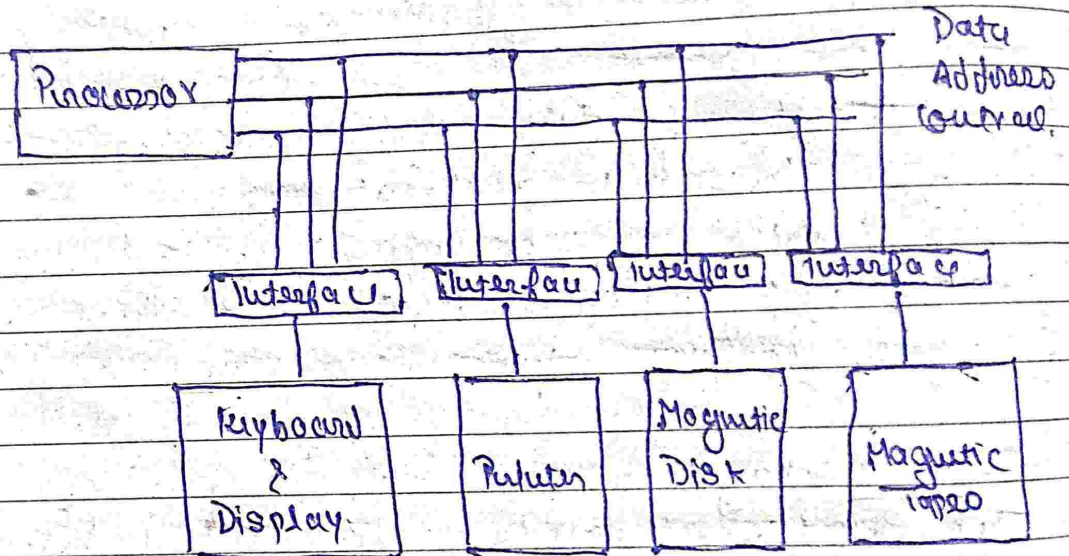
Discuss the design of typical I/O interface.

Sol

I/O interface :- Provide a method for transferring information between internal storage (Memory & registers) & External I/O devices

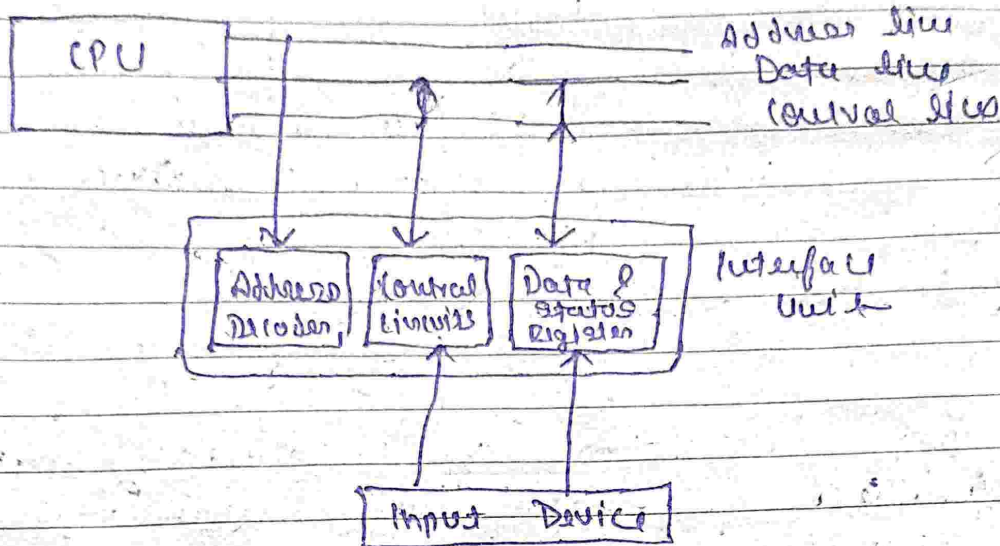


Q I/O devices & CPU cannot communicate directly



Connection of I/O bus to I/O interface

I/O interface of an Input Device



① Address Decoder :- Enables device to recognise its address when the address appears on the address line.

② Control Circuits :- Synchronises data flow and supervises the data transfer between processor and peripheral.

③ Data register :- hold data.

④ Status Register :- store status information.

⇒ 4 types of I/O commands :-

Instruction executed in interface & its attached peripheral services.

① Control command :- used to activate the peripheral & to tell it what to do.

② Status command :- used to test various status condition in the interface & the peripherals.

- ③ Output data :- Causes the interface to respond by transferring data from bus into one of its registers.
- ④ Input data :- Interface receives an input of data from the peripheral & places it in its buffer registers.

Q What is DMA? Describe how DMA is used to transfer data from peripherals directly from I/O.

A Why are the read and write control lines in a DMA controller bidirectional? Sol because the microprocessor fetch (read) the ~~data~~ data from the memory and write data to the memory.

Ques 2 Explain the difference between vectored and non-vectored interrupt. Explain stating example of each.

Sol Vectored Interrupt:-

- ① Vectored interrupt are those interrupt that generates the interrupt request, identifies itself directly to the processor.
- ② Vector interrupt have fixed memory location for transfer of control for normal execution.
- ③ Vectored interrupt has memory address.
- ④ Response time is low.

⑤ The vectored interrupt allows the CPU to be able to know what ISR to carry out in software.

⑥ TRAP is vectored interrupt.

Non Vectored Interrupts -

① Non-vectored interrupt are those in which vector address is not pre-defined.

② Non-vectored interrupt do not have fixed memory location for transfer of control for normal execution.

③ A non vectored interrupt do not have memory address

④ When a non-vectored interrupt received, it jump into the program counter to fixed address in hardware.

⑤ Response time is high

⑥ INTR is non-vectored interrupt

Q Diff Processor and IOP

Processor	IOP
1. Processor is CPU	1. IOP is part of the processing
2. Handles arithmetic and logical tasks.	2. Handles only I/O processing.
3. DMA controller is set up by CPU	3. IOP is a processor with DMA.

(ii) Synchronous and asynchronous transmission

Synchronous transmission	Asynchronous transmission
1. Transmitter and receiver are synchronized by clock	1. Transmitter and receiver are not synchronized by clock
2. Data bits are transmitted with synchronization of clock.	2. Bits of data are transmitted at constant rate
3. Data transfers takes place in blocks.	3. Data transfer is character oriented

(iv) character - oriented and Bit oriented protocols

character - oriented	Bit - oriented
1. The character - oriented protocol is based on the binary code of a character set.	1. The bit - oriented protocol does not use characters in its control field and is independent of any particular code

2. The code has 128 characters, of which 95 are graphics character and 33 are control character.

2. It allows the transmission of serial bit stream of any length without the implication of character boundaries.

(v) DMA

1. As DMA initialised, CPU become idle.

Interrupt initiated (I/O)

① CPU executes the current program, during the interrupt initiated I/O technique.

2. As the DMA disabled, memory buses are returned to CPU and CPU starts executing its program.

② After the transfers, CPU returns to the previous program to continue.

Q Write a note on Subroutine :- ~~by~~

Sol Subroutine :- A subroutine (function) is a subprogram in the main program that perform a specific task.

① During the execution of main program, a subroutine may be called at various points in the main program.

② A call subroutine instruction consist of an operation code along with an address that specify the beginning address of a subroutine.

③ The instruction is executed by performing two operations.

→ The address of next instruction (PC) is stored in register stack.

→ The PC is loaded with the starting address of the subroutine.

④ The last instruction from subroutine causes a return to the address stored in stack.

Subroutine call

(List of Micro-operations)

$SP \leftarrow SP - 1$

$PC \leftarrow PC$

$PC \leftarrow$ starting address of subroutine

Return from subroutine :- (List of micro-operations)

$PC \leftarrow PC$

$SP \leftarrow SP + 1$

Recursive Subroutine :- A recursive subroutine is a subroutine that itself again and again.

Q Difference between serial and parallel communication.

⇒ Basis for comparison	Serial Communication	Parallel Communication
→ Meaning	Data flows in bi-direction bit by bit	Multiple lines are used to send data i.e. 8 bits or 1 byte at a time
→ Cost	Economical	Expensive
→ Bits transferred at 1 clock pulse	1 bit	8 bits or 1 byte
→ Speed	Slow	Fast
→ Applications	Used for long distances communication, for example, computer to computer	Short distance, for example, computer to printer

Q I/O processor. IOP
Ans Already done